Table 1: The four scenarios as modifications of DDM/MDT.

Scenario	fixed / learned	objective	change vs DDM	change vs MDT
1. Accuracy (neural SVD)	$\mathbf{W}$ fixed; learn f_θ	$\min_\theta \lVert S f_\theta f_\theta^\top S - \mathbf{G}\rVert_F^2$	$\mathbf{A}^{2t} \to \mathbf{A}\mathbf{A}^\top$; multi-view input	SVD readout $\to$ NN learns it
2. Cost (truncated SVD)	$\mathbf{W}$ fixed; no network	top-k `svds` of $x \mapsto \mathbf{W}x$	drop the network	dense $O(N^3) \to$ matrix-free
3. Out-of-sample	$\mathbf{W}$ fixed; learn $x_* \mapsto \psi$	$f_\theta(x_*)$ vs Nyström $\mathbf{w}_*^\top \mathbf{V}$	use f_θ for new points	add extension (none in MDT)
4. End-to-end	learn ϕ_v, a: $\mathbf{W}(\theta)$	$\min_\theta \mathcal{L}_{\mathrm{contr}}(\mathbf{W}(\theta))$	$\mathbf{G} \to$ trainable $\mathbf{G}(\theta)$	operator fixed $\to$ learned

Setup. The map for the whole paper: each experiment is one edit to the DDM/MDT pipeline. $\mathbf{W}$ is the MDT operator, $S = \mathrm{diag}(\sqrt{\boldsymbol{\pi}})$ its $\sqrt{\boldsymbol{\pi}}$ weighting, $\mathbf{G}$ the DDM Gram target, $\mathbf{w}_*$ the operator row of a new point x_* against the training set. Scenario 1 (Accuracy): freeze $\mathbf{W}$, replace its SVD readout with a net f_θ trained by $\sqrt{\boldsymbol{\pi}}$-Gram matching. Scenario 2 (Cost): freeze $\mathbf{W}$, drop the net, take a matrix-free truncated `svds`. Scenario 3 (OOS): freeze $\mathbf{W}$, learn a map for new points and compare it to Nyström. Scenario 4 (End-to-end): make $\mathbf{W}(\theta)$ itself trainable under a contrastive loss.

Result. Scenarios 1–3 keep $\mathbf{W}$ fixed and only change how its SVD embedding is read out or extended, and all succeed or tie the classical baseline. Scenario 4 is the only one that removes the fixed SVD target, and it collapses. Every later table instantiates one of these four rows.

Dataset	classes	features only	eig (random)	$\mathrm{eig}_{\mathrm{best}}$ (fair)	$\mathrm{eig}_{\mathrm{best}}$ (oracle)	neural (fair)
MNIST-4	4	0.707	0.789	0.783	0.808	0.782
MNIST-10k	10	0.450	0.786	0.787	0.798	0.790
ALOI	100	0.737	0.773	0.767	0.787	0.795

Setup. Three clustering benchmarks, 4,000 samples each, scored by k-means AMI: MNIST-4 (digits 0–3, 4 classes), MNIST-10k (all digits, 10 classes), ALOI (100-object subset of the Amsterdam Library of Object Images). MDT embeds by plain truncated `svds` of its operator. Here k is the number of clusters (the known class count, which k-means also needs), and the embedding is the top-k singular vectors of the operator after dropping the trivial top mode (σ_1, the constant stationary direction), scaled by their singular values; by Eckart–Young this is the optimal rank-k summary, and we keep exactly k because the clustering that follows asks for k groups (extra components only add noise on the near-uniform diffused operator). All columns share this decomposition: features only: k-means on raw features, no diffusion; eig (random): mean over 8 random trajectories; $\mathrm{eig}_{\mathrm{best}}$ (fair): best of 8 selected by top-k spectral energy (unsupervised, deployable); $\mathrm{eig}_{\mathrm{best}}$ (oracle): best of 8 selected by AMI (uses labels); neural (fair): a net reproducing the top-k SVD of the same energy-selected (fair) operator, by a Rayleigh objective (the Rayleigh trace $\mathrm{tr}(Q^\top \mathbf{W}\mathbf{W}^\top Q)$ is a differentiable surrogate whose optimum equals the SVD, so gradient ascent learns the SVD as a function, whereas an eigensolver only computes it as numbers).

Result. With the decomposition fixed to `svds`, trajectory selection adds nothing (eig random $\approx$ $\mathrm{eig}_{\mathrm{best}}$ fair); the oracle leads only by selecting on labels (AMI), so it is not deployable. Neural matches, and marginally exceeds, `svds` (ALOI +0.03), though within run-to-run noise: a hint, not yet a result, that a learned approximation may leave small headroom over MDT.

Table 2: Scaling on MNIST-10k (Scenario 2).

	wall-clock time			accuracy
N	dense `eigh`	`svds`	neural	AMI (svds $\approx$ neural)
2 000	4.3 s	0.07 s	18.5 s	0.74
4 000	14.6 s	0.11 s	40.0 s	0.79
8 000	98.0 s	0.30 s	71.3 s	0.81
10 000	n/a	0.31 s	89.4 s	0.81

Setup. Time to build the embedding of N points three ways, plus the resulting k-means AMI. dense `eigh` = classical $O(N^3)$ eigendecomposition of the dense operator; `svds` = truncated top-k SVD of the sparse, matrix-free operator action (Scenario 2); neural = the Rayleigh-trained net (Scenario 1). `svds` and neural produce the same embedding, hence one shared accuracy column; dense `eigh` is omitted at N=10k (too slow).

Result. `svds` (0.07–0.31 s) is ~300× faster than dense `eigh` and ~200× faster than the net, at identical accuracy (0.74–0.81). The scalable classical method wins; the neural machinery buys no speed.

Table 3: Out-of-sample AMI over 5 seeds (Scenario 3).

Dataset	Nyström	deep ($\sqrt{\boldsymbol{\pi}}$-Gram)
MSRC-v5	0.685±0.021	0.648±0.093
Handwritten	0.859±0.028	0.866±0.030
Wikipedia	0.307±0.102	0.295±0.057
UCI	0.801±0.054	0.826±0.053
OutdoorScene	0.490±0.041	0.480±0.040
Caltech101-7	0.525±0.044	0.527±0.043

Setup. A held-out test split is embedded and clustered by two out-of-sample extensions of the same fixed operator $\mathbf{W}$ (mean±std over 5 seeds; OutdoorScene 4, one seed dropped to a degenerate view). Nyström = the SVD's own training-free extension: form the new point's operator row $\mathbf{w}_*$ against the training set and project it onto the training singular vectors, $\psi(x_*) = \mathbf{w}_*^\top \mathbf{V}$. deep ($\sqrt{\boldsymbol{\pi}}$-Gram) = the DDM encoder f_θ (one MLP branch per view), the same encoder for every dataset, with no feature/affinity variants. It is trained by $\sqrt{\boldsymbol{\pi}}$-Gram matching: let $\boldsymbol{\pi}$ be the stationary distribution of $\mathbf{W}$ ($\boldsymbol{\pi}^\top \mathbf{W} = \boldsymbol{\pi}^\top$), $\mathbf{A} = \mathbf{\Pi}^{1/2}\mathbf{W}\mathbf{\Pi}^{-1/2}$ its $\sqrt{\boldsymbol{\pi}}$-symmetrised form, and $\mathbf{G} = \mathbf{A}\mathbf{A}^\top - \sqrt{\boldsymbol{\pi}}\sqrt{\boldsymbol{\pi}}^\top$ the target Gram (with the trivial top mode $\sqrt{\boldsymbol{\pi}}$ removed). The loss makes the encoder's $\sqrt{\boldsymbol{\pi}}$-scaled Gram match it, $\min_\theta \left\|(\sqrt{\boldsymbol{\pi}}\odot F)(\sqrt{\boldsymbol{\pi}}\odot F)^\top - \mathbf{G}\right\|_F^2$, $F = f_\theta(X)$. By Eckart–Young its optimum is the top-k eigenspace of $\mathbf{G}$ (the diffusion-map embedding), so the net learns that embedding as a function and extends to new points by a forward pass. The $\sqrt{\boldsymbol{\pi}}$ scaling is essential, not cosmetic: a constant (collapsed) embedding lands exactly on the removed $\sqrt{\boldsymbol{\pi}}\sqrt{\boldsymbol{\pi}}^\top$ mode, so it is penalised; drop it and the encoder collapses to a constant (verified here: AMI $\to$ 0.02).
Result. Over 5 seeds Nyström and the encoder tie within one std on every dataset (row gaps ≤ 0.04, inside the $\pm$0.02–0.10 spreads): out-of-sample is a wash. The single-run gaps that once looked decisive are seed noise. A faithfully-trained encoder does not beat the SVD's free Nyström extension; at best it reproduces it.

Table 4: End-to-end learning of the geometry fails (Scenario 4).

Dataset	fixed+random	fixed+learned wts	**learned geom.**
MSRC-v5	0.682	0.470	0.328
Caltech101-7	0.535	0.410	**0.049**
OutdoorScene	0.570	0.463	**0.005**
UCI	0.822	0.914	0.751

Setup. AMI as progressively more of the operator is made trainable (left$\to$right). fixed+random: the per-view kernels P_v (Gaussian kNN affinities) and the fusion are both fixed, a random convex trajectory, no learning. fixed+learned wts: keep the kernels P_v fixed and learn only the mixing weights, the $t \times V$ matrix a (one softmax over views per step) that convexly combines the fixed per-view walks, $\mathbf{W} = \prod_s \left(\sum_v a_{sv} P_v\right)$, by Adam on the contrastive loss (positive pairs = kernel neighbours). learned geom.: also learn the geometry itself: a per-view feature map ϕ_v defines each kernel $P_v(\phi_v)$, so both the kernels and a are trained end-to-end under the same loss (now even the Gram target moves). Four datasets; bold = near-zero collapse.
Result. Accuracy falls monotonically as more is learned: fixed > learned-weights > fully-learned (near-zero on Caltech/OutdoorScene). Learning the operator hurts. The mechanism is optimisation/conditioning (flat loss, spectrum flattening), not the literal rank-one collapse the numbers suggest.

Table 5: t-sweep of three methods on the four collapse datasets (AMI).

Dataset	method	t=1	t=2	t=3	t=4
MSRC-v5	fixed MDT	.704	.736	.728	.720
	imit. (Gram)	.712	.724	.701	.706
	end-to-end	.533	.309	.277	.239
Caltech101-7	fixed MDT	.547	.588	.619	.620
	imit. (Gram)	.643	.545	.647	.644
	end-to-end	.234	.094	.070	.085
OutdoorScene	fixed MDT	.629	.633	.606	.598
	imit. (Gram)	.579	.561	.522	.525
	end-to-end	.340	.001	.098	.177
UCI	fixed MDT	.846	.834	.830	.840
	imit. (Gram)	.841	.810	.840	.828
	end-to-end	.867	.837	.763	.730

Setup. For diffusion length t=1..4, three ways to get the embedding. fixed MDT = best-of-8 length-t operator, plain SVD (no learning); imit. (Gram) = a $\sqrt{\boldsymbol{\pi}}$-Gram encoder trained to reproduce that fixed operator; end-to-end = learn the operator under the contrastive loss (Scenario 4, 5-seed mean). Four datasets.
Result. Fixed and imit are stable across t and coincide (imit reproduces the fixed operator at every length), while end-to-end collapses as t grows (well-connected kNN graphs over-mix past one hop), except UCI (one dominant view).
Why imit beats end-to-end. Two reasons. (i) Fixed vs. moving target: imit matches the Gram of a fixed, known-good operator (the best-of-8 MDT), so its target never drifts; end-to-end instead learns the operator, so its target $\mathbf{G}(\theta)$ moves and slides into a degenerate, spectrum-flattened optimum. (ii) Conditioning: Gram matching is a well-behaved quadratic, $\|FF^\top - \mathbf{G}\|^2$ with gradient $4(FF^\top - \mathbf{G})F$, so it descends cleanly, whereas the contrastive loss saturates once the operator's rows are near-uniform (its exp/softmax gives tiny, near-equal gradients) and stalls. imit therefore reaches higher AMI because it inherits the good fixed operator intact, while end-to-end degrades that operator as it trains.

Table 6: Unsupervised MDT trajectory selection (Scenario 4).

	Spearman ρ			selection regret		
Dataset	Q_c	sil	**en**	Q_c	sil	**en**
MSRC-v5	−.25	−.17	**.49**	.072	.097	**.043**
Yale	−.12	−.59	.11	.073	.131	**.064**
BBCSport	.13	−.04	**.60**	.125	.164	**.019**
Caltech101-7	−.12	.11	**.36**	.047	.467	**.043**
Handwritten	.24	.05	.19	**.050**	.365	.209
UCI	−.28	−.30	**.36**	.369	.404	**.121**
Wikipedia-test	.33	.61	−.47	**.000**	**.000**	.315
mean	−.01	−.05	**.23**	.105	.233	.116

Setup. Can an unsupervised index pick the best trajectory? Over 30 random trajectories ×3 seeds on structured datasets (oracle AMI$\geq$ 0.35), rank trajectories by three indices and score each two ways: Spearman ρ with the oracle-AMI ranking (higher better) and selection regret = oracle−selected AMI (lower better). Indices: Q_c = MDT's contrastive Internal Quality, sil = embedding silhouette, en = spectral energy (top-k explained variance). Embedding dim d=k; bold = best per block.

Result. Spectral energy is the only positive ranker (mean ρ +.23 vs −.01, −.05) and lowest-regret on 5/7, but it is weak and fails on heterogeneous-view Wikipedia-test (ρ −.47; four more cross-modal cases below). Raising d (64/128) inflates ρ/regret yet lowers deployed AMI (0.54→0.52), a flat-spectrum mirage, not a gain. No index selects reliably.

Why energy fails on heterogeneous views. Energy ranks trajectories by spectral cleanliness, not cluster correctness. On cross-modal data a view can be internally clean (high energy) yet uninformative about the labels (low single-view AMI), a clean-but-useless view; a trajectory that leans on it keeps high energy but clusters poorly, so energy anti-correlates with AMI. A separate probe of five heterogeneous datasets (per-view energy vs. single-view AMI, and selector ρ over 25 trajectories ×3 seeds) shows energy fails exactly when such a view is present:

Dataset	views	energy ρ	clean-but-useless view (en/AMI)	verdict
Cora	4	−.61	v2, v3: .14/.00	fails
Wikipedia	2	−.59	v0: .07/.04	fails
Prokaryotic	3	−.38	v2: .11/.04	fails
NUS-WIDE	5	−.34	v3: .06/.04	fails
CiteSeer	2	+.93	none	works

Energy fails ($\rho < 0$) on the four datasets carrying a clean-but-useless view and works (ρ +.93) on the one without. Cora is the sharpest: its two highest-energy views (.14) have AMI ≈.00, while the most informative view (AMI .13) has the lowest energy (.01), so energy points the wrong way (energy-vs-AMI across views $\rho = -.95$). Energy measures clean structure, not correct structure, so it cannot detect a clean-but-useless view: the failure is fundamental, not a tuning issue.